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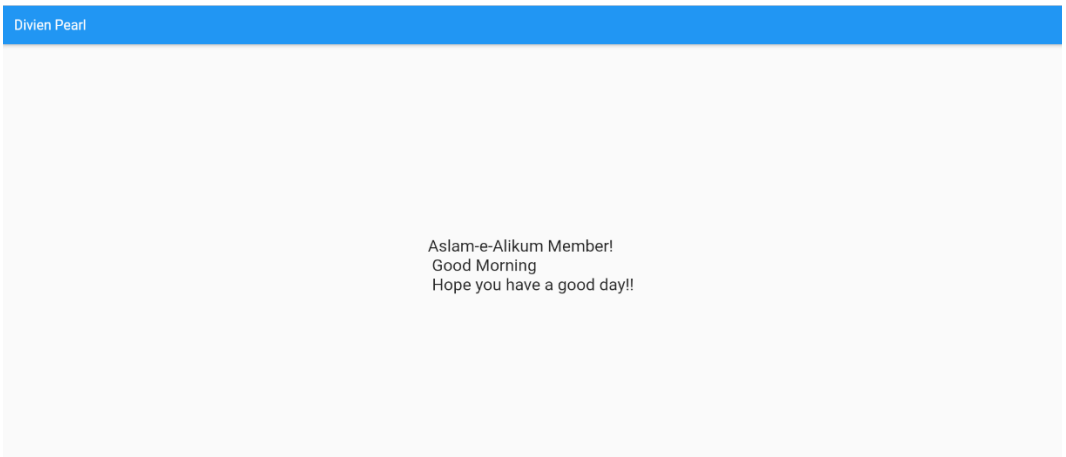
Task 01: Run on Chrome Add Screenshot.

main.dart

lib > main.dart

```dart
import 'package:flutter/material.dart';
Run Application
void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        appBar: AppBar(
          title: Text("Divien Pearl"),
        ), // AppBar
        body: Center(
          child: Text(
            "Aslam-e-Alikum Member! \n Good Morning \n Hope you have a good day!! "
            style: TextStyle(fontSize: 24),
          ), // Text
        ), // Center
      ), // Scaffold
    ); // MaterialApp
  }
}
```



main.dart

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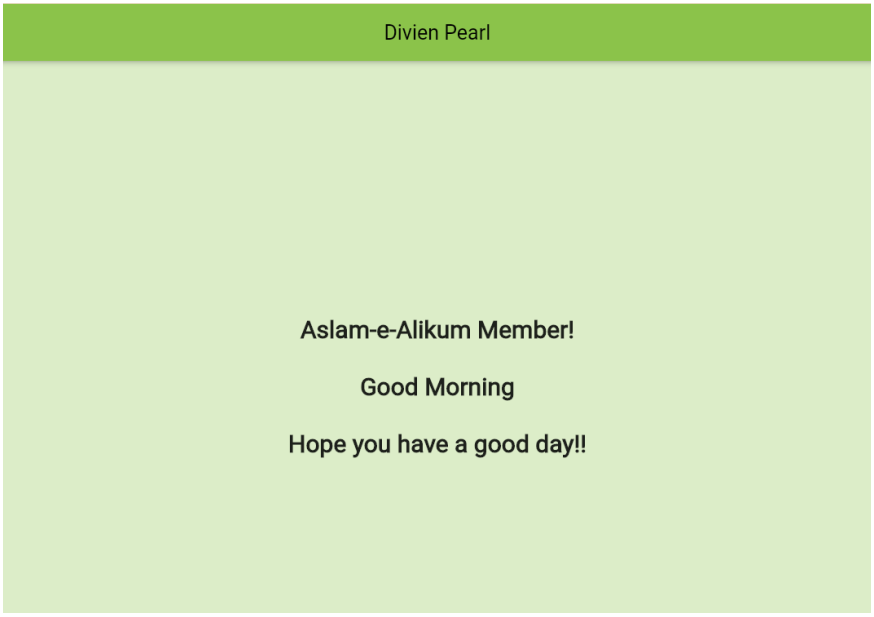
```dart
import 'package:flutter/material.dart';
Run Application
void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
```

main.dart

lib > main.dart

```dart
      theme: ThemeData(
        primarySwatch: Colors.lightGreen,
        canvasColor: Colors.lightGreen.shade100,
        platform: TargetPlatform.iOS,
      ), // ThemeData

      home: Scaffold(
        appBar: AppBar(
          title: Text("Divien Pearl"),
        ), // AppBar
        body: Center(
          child: Text(
            "Aslam-e-Alikum Member! \n\nGood Morning \n\nHope you have a good day!!",
            textAlign: TextAlign.center,   // Center text properly
            style: TextStyle(
              fontSize: 24,
              fontWeight: FontWeight.bold,
              color: Colors.black87,
            ), // TextStyle
          ), // Text
        ), // Center
      ), // Scaffold
    ); // MaterialApp
  }
}
```



main.dart ×

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```dart
import 'package:flutter/material.dart';
Run Application
void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,

      // Global Theme
      theme: ThemeData(
        primarySwatch: Colors.lightGreen,
        canvasColor: Colors.lightGreen.shade100,
      ), // ThemeData

      home: HomeScreen(),
    ); // MaterialApp
  }
}
// ================= HOME SCREEN =================

class HomeScreen extends StatefulWidget {
```

main.dart ×

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```dart
  @override
  _HomeScreenState createState() => _HomeScreenState();
}
class _HomeScreenState extends State<HomeScreen> {

  int _counter = 0;
  void _incrementCounter() {
    setState(() {
      _counter++;
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Flutter Lab 3"),
      ), // AppBar

      body: SingleChildScrollView(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [

            SizedBox(height: 20),
```

main.dart

lib > main.dart

```dart
// ===== STATELESS WIDGET =====
Instructions(),

SizedBox(height: 20),

// ===== THEME OVERWRITE =====
Theme(
  data: ThemeData(
    cardColor: Colors.deepOrange,
  ), // ThemeData
  child: Card(
    margin: EdgeInsets.all(15),
    child: Padding(
      padding: EdgeInsets.all(15),
      child: Text(
        "Unique ThemeData (Overwrite)",
        style: TextStyle(fontSize: 18),
      ), // Text
    ), // Padding
  ), // Card
), // Theme

// ===== THEME COPYWITH =====
Theme(
  data: Theme.of(context)
```

main.dart

lib > main.dart

```dart
    .copyWith(cardColor: Colors.deepOrange),
  child: Card(
    margin: EdgeInsets.all(15),
    child: Padding(
      padding: EdgeInsets.all(15),
      child: Text(
        "copyWith Theme (Extended)",
        style: TextStyle(fontSize: 18),
      ), // Text
    ), // Padding
  ), // Card
), // Theme

SizedBox(height: 20),

// ===== STATEFUL WIDGET (MAXIMUM BID) =====
MaximumBid(),

SizedBox(height: 30),

// ===== COUNTER DISPLAY =====
Text(
  "Counter Value: $_counter",
  style: TextStyle(
    fontSize: 22,
```

main.dart

lib > main.dart

```
                fontWeight: FontWeight.bold,
              ), // TextStyle
            ), // Text

            SizedBox(height: 30),
          ],
        ), // Column
      ), // SingleChildScrollView

      // ===== FLOATING ACTION BUTTON =====
      floatingActionButton: FloatingActionButton(
        onPressed: _incrementCounter,
        child: Icon(Icons.add),
      ), // FloatingActionButton
    ); // Scaffold
  }
}

// ================= STATELESS WIDGET =================

class Instructions extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Text(
      'StatelessWidget:\nData does NOT change.',
```

main.dart

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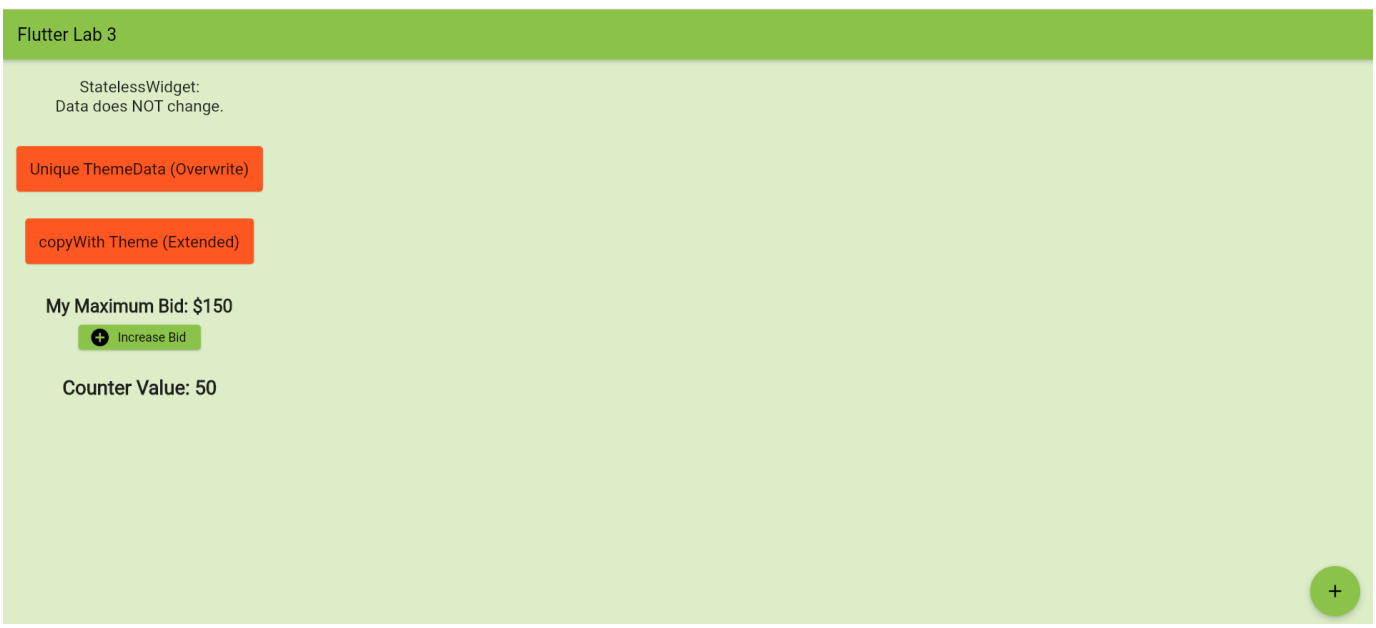
```
      textAlign: TextAlign.center,
      style: TextStyle(
        fontSize: 18,
        fontWeight: FontWeight.w500,
      ), // TextStyle
    ); // Text
  }
}

// ================ STATEFUL WIDGET ================

class MaximumBid extends StatefulWidget {
  @override
  _MaximumBidState createState() => _MaximumBidState();
}

class _MaximumBidState extends State<MaximumBid> {

  double _maxBid = 0.0;

  void _increaseMyMaxBid() {
    setState(() {
      _maxBid += 50.0;
    });
  }
```

main.dart

lib > main.dart

```dart
  @override
  Widget build(BuildContext context) {
    return Column(
      children: [
        Text(
          'My Maximum Bid: \$$_maxBid',
          style: TextStyle(
            fontSize: 20,
            fontWeight: FontWeight.bold,
          ), // TextStyle
        ), // Text
        SizedBox(height: 10),
        ElevatedButton.icon(
          onPressed: _increaseMyMaxBid,
          icon: Icon(Icons.add_circle),
          label: Text('Increase Bid'),
        ), // ElevatedButton.icon
      ],
    ); // Column
  }
}
```



Task 02: Run on Emulator and Add Screenshot.

main.dart

```dart
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,

      // Global Theme
      theme: ThemeData(
        primarySwatch: Colors.lightGreen,
        canvasColor: Colors.lightGreen.shade100,
      ), ThemeData

      home: HomeScreen(),
    ); MaterialApp
  }
```

```dart
}
// ================ HOME SCREEN ================

class HomeScreen extends StatefulWidget {
  @override
  _HomeScreenState createState() => _HomeScreenState();
}
class _HomeScreenState extends State<HomeScreen> {

  int _counter = 0;
  void _incrementCounter() {
    setState(() {
      _counter++;
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
```

main.dart

```dart
      title: Text("Flutter Lab 3"),
    ), AppBar

    body: SingleChildScrollView(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [

          SizedBox(height: 20),

          // ===== STATELESS WIDGET =====
          Instructions(),

          SizedBox(height: 20),

          // ===== THEME OVERWRITE =====
          Theme(
            data: ThemeData(
              cardColor: Colors.deepOrange,
            ), ThemeData
            child: Card(
```

main.dart

```dart
              margin: EdgeInsets.all(15),
              child: Padding(
                padding: EdgeInsets.all(15),
                child: Text(
                  "Unique ThemeData (Overwrite)",
                  style: TextStyle(fontSize: 18),
                ), Text
              ), Padding
            ), Card
          ), Theme

          // ===== THEME COPYWITH =====
          Theme(
            data: Theme.of(context)
                .copyWith(cardColor: Colors.deepOrange),
            child: Card(
              margin: EdgeInsets.all(15),
              child: Padding(
                padding: EdgeInsets.all(15),
                child: Text(
                  "copyWith Theme (Extended)",
```

main.dart

```
                    style: TextStyle(fontSize: 18),
                  ), Text
                ), Padding
              ), Card
            ), Theme

            SizedBox(height: 20),

            // ===== STATEFUL WIDGET (MAXIMUM BID) =====
            MaximumBid(),

            SizedBox(height: 30),

            // ===== COUNTER DISPLAY =====
            Text(
              "Counter Value: $_counter",
              style: TextStyle(
                fontSize: 22,
                fontWeight: FontWeight.bold,
              ), TextStyle
            ), Text
```

main.dart

```
            SizedBox(height: 30),
          ],
        ), Column
      ), SingleChildScrollView

      // ===== FLOATING ACTION BUTTON =====
      floatingActionButton: FloatingActionButton(
        onPressed: _incrementCounter,
        child: Icon(Icons.add),
      ), FloatingActionButton
    ); Scaffold
  }
}

// ================ STATELESS WIDGET ================

class Instructions extends StatelessWidget {
  const Instructions({super.key});

  @override
```

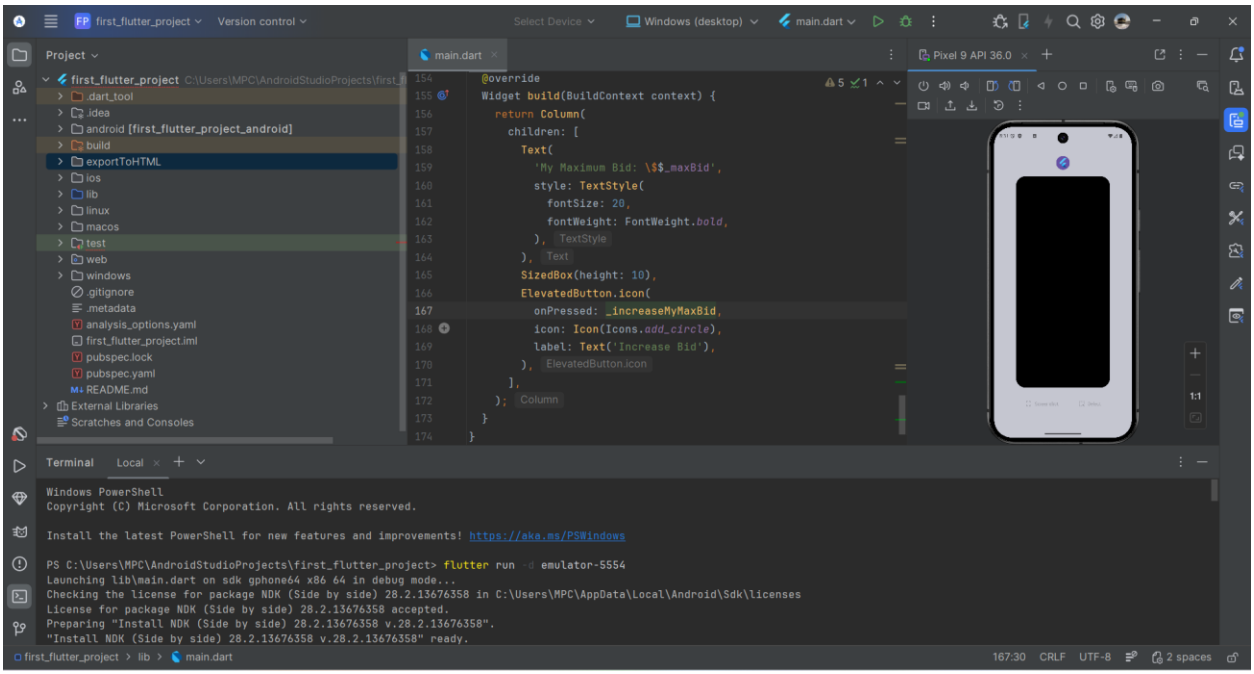
main.dart

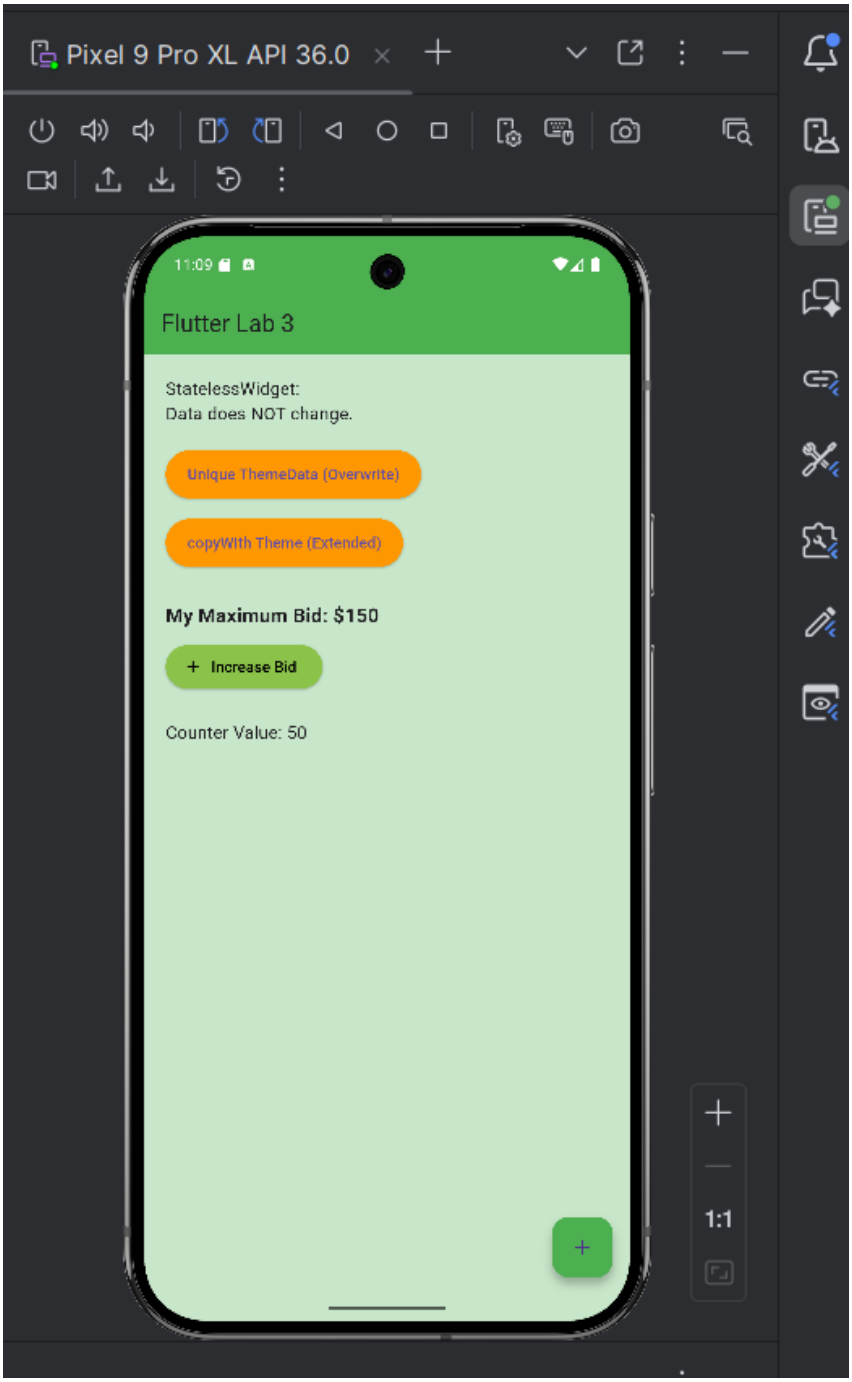
```dart
  Widget build(BuildContext context) {
    return Text(
      'StatelessWidget:\nData does NOT change.',
      textAlign: TextAlign.center,
      style: TextStyle(
        fontSize: 18,
        fontWeight: FontWeight.w500,
      ),
    );
  }
}

// ================= STATEFUL WIDGET =================

class MaximumBid extends StatefulWidget {
  @override
  _MaximumBidState createState() => _MaximumBidState();
}

class _MaximumBidState extends State<MaximumBid> {
```

main.dart

```dart
  double _maxBid = 0.0;

  void _increaseMyMaxBid() {
    setState(() {
      _maxBid += 50.0;
    });
  }

  @override
  Widget build(BuildContext context) {
    return Column(
      children: [
        Text(
          'My Maximum Bid: \$$_maxBid',
          style: TextStyle(
            fontSize: 20,
            fontWeight: FontWeight.bold,
          ),
        ),
        SizedBox(height: 10),
        ElevatedButton.icon(
```

```
167            onPressed: _increaseMyMaxBid,
168            icon: Icon(Icons.add_circle),
169            label: Text('Increase Bid'),
170          ),  ElevatedButton.icon
171        ],
172      );  Column
173    }
174  }
```




Pixel 9 Pro XL API 36.0
11:09
Flutter Lab 3
StatelessWidget:
Data does NOT change.
Unique ThemeData (Overwrite)
copyWith Theme (Extended)
My Maximum Bid: $150
+ Increase Bid
Counter Value: 50
1:1